

Due: August 18, 2026

1. Given $q = \frac{(v+2)^3-8}{v}$ and ($v \neq 0$), find:

a) $\lim_{v \rightarrow 0} q$

b) $\lim_{v \rightarrow 2} q$

2. Compute the following limits:

a) $\lim_{x \rightarrow \infty} \frac{x}{x^2 + 1}$

b) $\lim_{x \rightarrow \infty} \frac{x^2}{x^2 + 1}$

c) $\lim_{x \rightarrow \infty} \frac{x^4}{x^2 + 1}$

d) $\lim_{x \rightarrow 8+} \frac{5}{x - 8}$

e) $\lim_{x \rightarrow 8-} \frac{-5}{x - 8}$

3. Given the function $y = 3x + 5$

a) Find the difference quotient as a function of $x + \Delta x$.

b) Find the derivative dy/dx .

c) Find $f'(1)$ and $f'(3)$.

4. Given the function $y = 5x^2 - 4x$

a) Find the difference quotient as a function of $x + \Delta x$.

b) Find the derivative dy/dx .

c) Find $f'(2)$ and $f'(3)$.

5. Compute the derivatives for the following functions by use of the power rule:

a) $f(x) = -6x^3 + 12x^2 - 4x + 7$

b) $f(x) = \frac{1}{10}x^{10} + \frac{1}{9}x^9 - \frac{1}{8}x^8$

c) $f(x) = (3x + 5)^2$ (expand first)

d) $f(x) = 100x^2 + \frac{1}{100}x$

6. Compute the derivatives y' two ways: product rule and expanding first:

a) $y = (x^2 + 1)(x^3 + 1)$

b) $y = (x - 1)(x^5 + x^4 + x^3 + x^2 + x + 1)$

c) $y = (x^3 + 1)(x^2 + 1)(x + 1)$

7. Compute the following derivatives by using the quotient rule:

a) $y = \frac{5x + 6}{x^2 + 1}$

b) $y = \frac{x^3 - x}{x^5 - x^3 + 1}$

8. Compute the following derivatives by using the quotient rule and the product rule:

a) $y = \frac{(x^3 - 2x^2 + 7x + 1)(x^2 - x + 3)}{x^3 + 1}$

b) $y = \frac{(x^2 + 1)(x^4 + 1)}{(x^3 + 1)(x^5 + 1)}$

9. Differentiate the following functions:

a) e^{5x}

b) $e^{\sqrt[3]{x}}$

c) $(e^e)^x$

d) 7^x

e) $2^{(x^2+6x-5)}$

f) $\ln(x^4 - x^2 + 1)$

g) $\ln(e^x)$

h) $\ln(2x)$